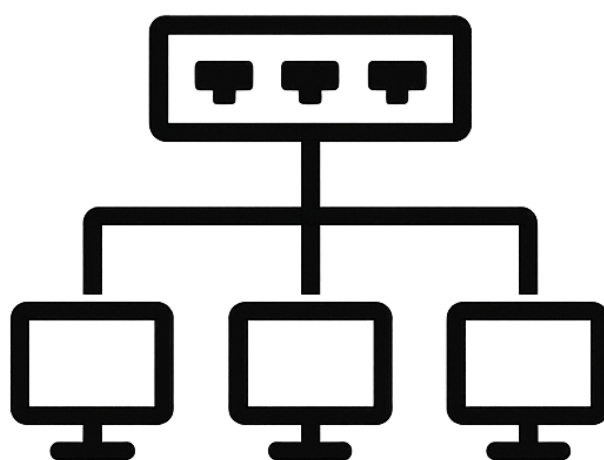


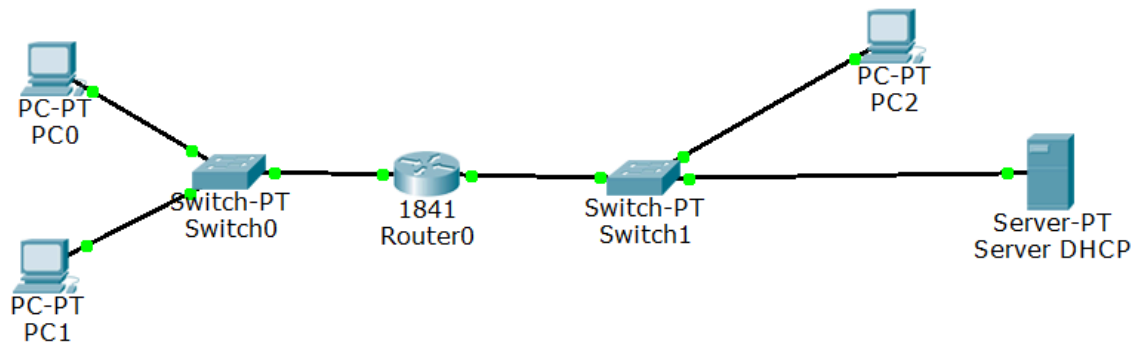


TP RÉSEAU

Objectifs :

Séparer le réseau en VLAN10 et VLAN20 et distribuer de IP avec un serveur DHCP

Topologie :



Server DHCP

Physical Config Desktop Custom Interface

GLOBAL

Settings

Algorithm Settings

SERVICES

HTTP

DHCP

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

FIREWALL

Pv6 FIREWALL

INTERFACE

FastEthernet0

DHCP

Service ☒ On ☐ Off

Pool Name

Default Gateway

DNS Server

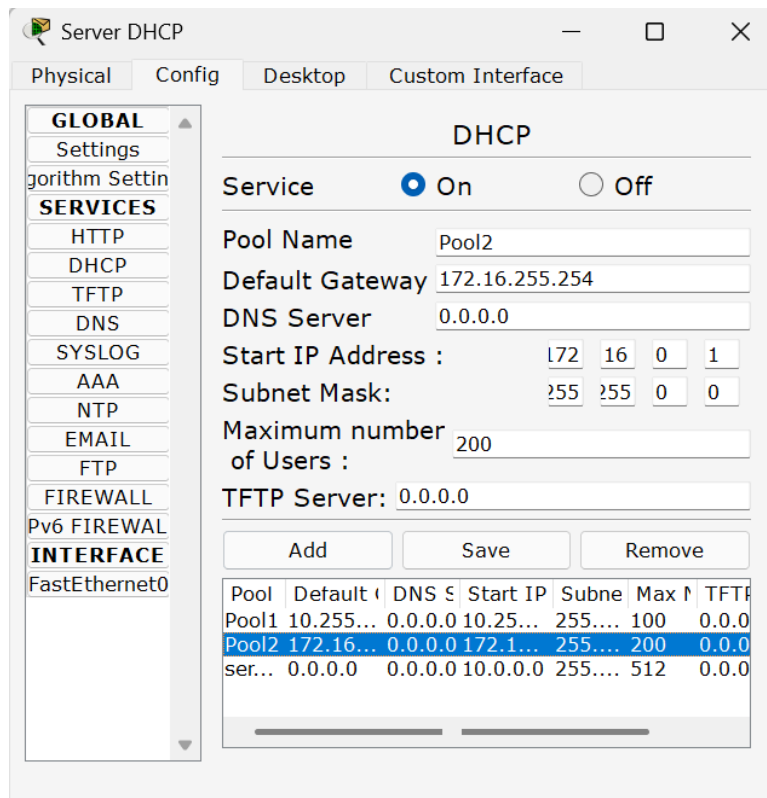
Start IP Address :

Subnet Mask:

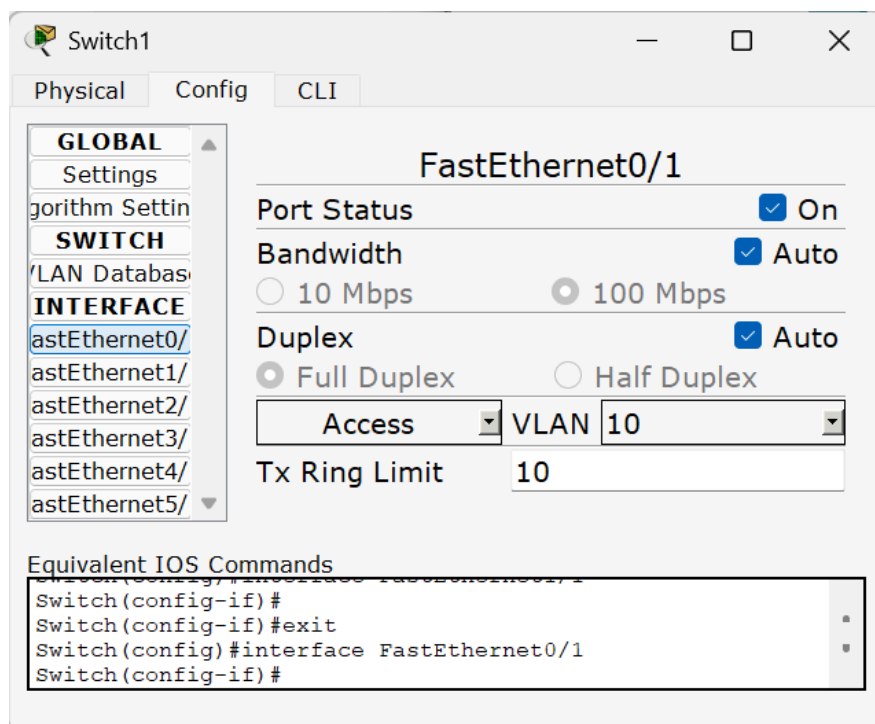
Maximum number of Users :

TFTP Server:

Pool	Default IP	DNS Server	Start IP	Subnet Mask	Max Users	TFTP Server
Pool1	10.255.255.254	0.0.0.0	10.255.255.1	255.255.255.0	100	0.0.0.0
Pool2	172.16.0.1	0.0.0.0	172.16.0.1	255.255.255.0	200	0.0.0.0
server	0.0.0.0	0.0.0.0	10.0.0.0	255.255.255.0	512	0.0.0.0



Le serveur DHCP est activé et a deux pools différents.



Switch1

Physical Config CLI

GLOBAL

Settings

Algorithm Settings

SWITCH

VLAN Database

INTERFACE

FastEthernet0/

FastEthernet1/

FastEthernet2/

FastEthernet3/

FastEthernet4/

FastEthernet5/

FastEthernet1/1

Port Status ☒ On

Bandwidth ☒ Auto

☐ 10 Mbps ☒ 100 Mbps

Duplex ☒ Auto

☒ Full Duplex ☐ Half Duplex

Access VLAN 10

Tx Ring Limit 10

Equivalent IOS Commands

```
Switch(config)#  
Switch(config-if)#  
Switch(config-if)#exit  
Switch(config)#interface FastEthernet1/1  
Switch(config-if)#
```

Switch0

Physical Config CLI

GLOBAL

Settings

Algorithm Settings

SWITCH

VLAN Database

INTERFACE

FastEthernet0/

FastEthernet1/

FastEthernet2/

FastEthernet3/

FastEthernet4/

FastEthernet5/

FastEthernet0/1

Port Status ☒ On

Bandwidth ☒ Auto

☐ 10 Mbps ☒ 100 Mbps

Duplex ☒ Auto

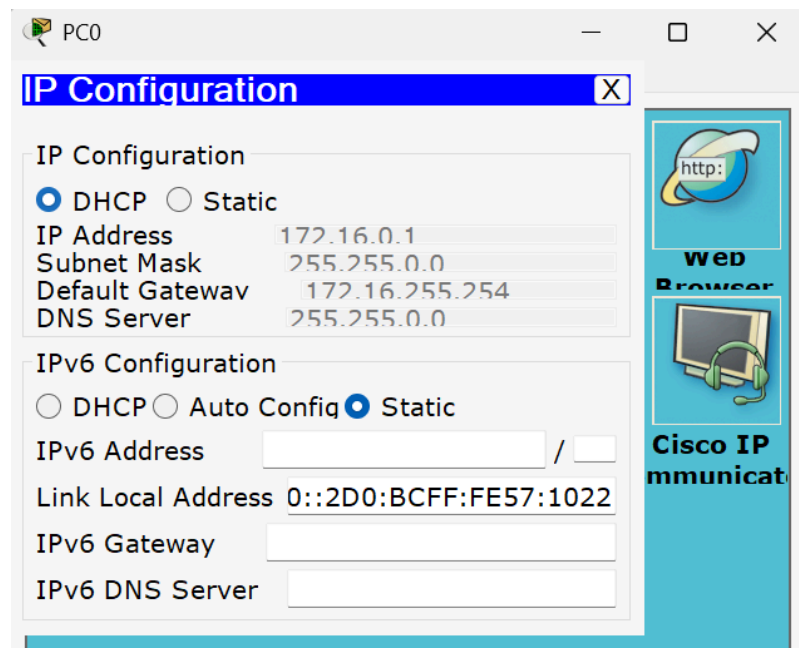
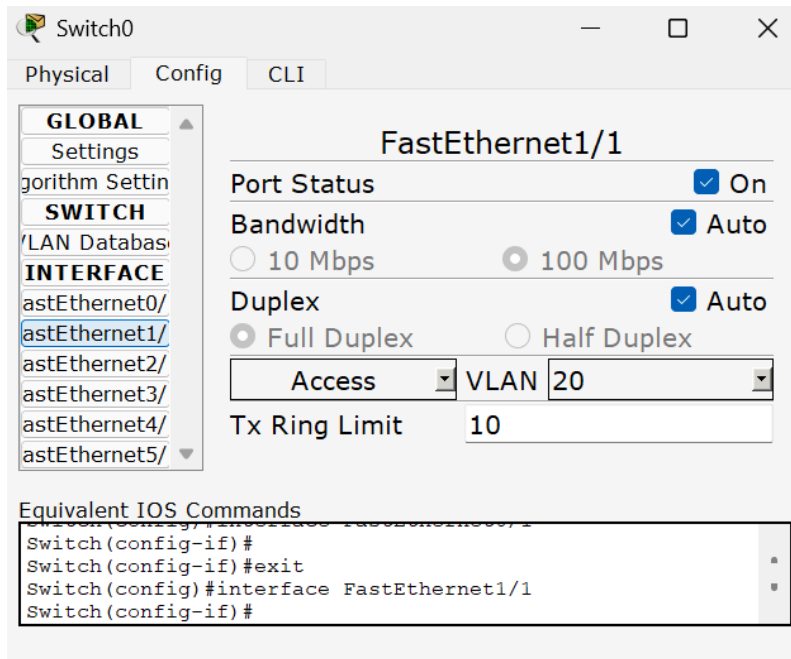
☒ Full Duplex ☐ Half Duplex

Access VLAN 20

Tx Ring Limit 10

Equivalent IOS Commands

```
Switch(config)#  
Switch(config-if)#  
Switch(config-if)#exit  
Switch(config)#interface FastEthernet0/1  
Switch(config-if)#
```



PC1

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 172.16.0.2

Subnet Mask 255.255.0.0

Default Gateway 172.16.255.254

DNS Server 255.255.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address 0::230:A3FF:FE07:B5E6

IPv6 Gateway

IPv6 DNS Server

Web Browser

Cisco IP Communicator

PC2

IP Configuration

IP Configuration

☒ DHCP ☐ Static

IP Address 10.255.255.1

Subnet Mask 255.0.0.0

Default Gateway 10.255.255.254

DNS Server 255.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address 0::2D0:58FF:FEA3:1C8D

IPv6 Gateway

IPv6 DNS Server

Web Browser

Cisco IP Communicator

Switch0

Physical
Config
CLI

GLOBAL
Settings
Algorithm Settings
SWITCH
VLAN Databases
INTERFACE
FastEthernet0/
FastEthernet1/
FastEthernet2/
FastEthernet3/
FastEthernet4/
FastEthernet5/

VLAN Configuration
VLAN Number
VLAN Name
Add
Remove

VLAN No	VLAN Name
1	default
10	VLAN10
11	VLAN11
20	VLAN20
1002	fddi-default
1003	token-ring-default

Equivalent IOS Commands

```
Switch(config)#interface FastEthernet1/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
```

Switch1

Physical
Config
CLI

GLOBAL
Settings
Algorithm Settings
SWITCH
VLAN Databases
INTERFACE
FastEthernet0/
FastEthernet1/
FastEthernet2/
FastEthernet3/
FastEthernet4/
FastEthernet5/

VLAN Configuration
VLAN Number
VLAN Name
Add
Remove

VLAN No	VLAN Name
1	default
10	VLAN10
20	VLAN20
21	VLAN21
1002	fddi-default
1003	token-ring-default

Equivalent IOS Commands

```
Switch(config)#interface FastEthernet1/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
```

Switch0

Physical Config CLI

IOS Command Line Interface

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa3/1, Fa4/1, Fa5/1
10	VLAN10	active	
11	VLAN11	active	
20	VLAN20	active	Fa0/1, Fa1/1, Fa2/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>show interfaces trunk
```

```
Switch>
```

Copy Paste

Switch1

Physical Config CLI

IOS Command Line Interface

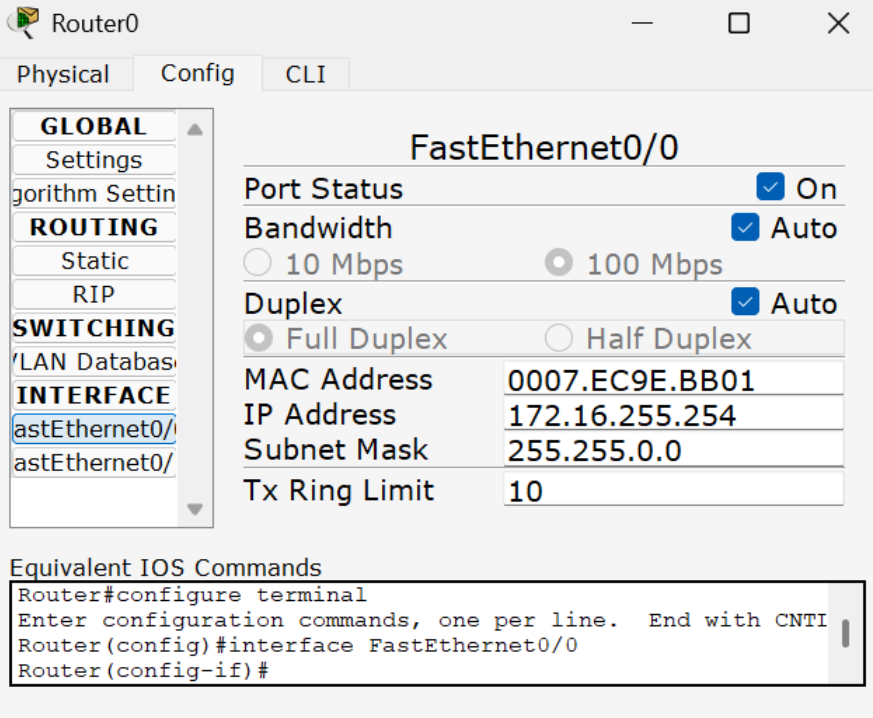
```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa3/1, Fa4/1, Fa5/1
10	VLAN10	active	Fa0/1, Fa1/1, Fa2/1
20	VLAN20	active	
21	VLAN21	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Copy Paste

configuration du routeur :

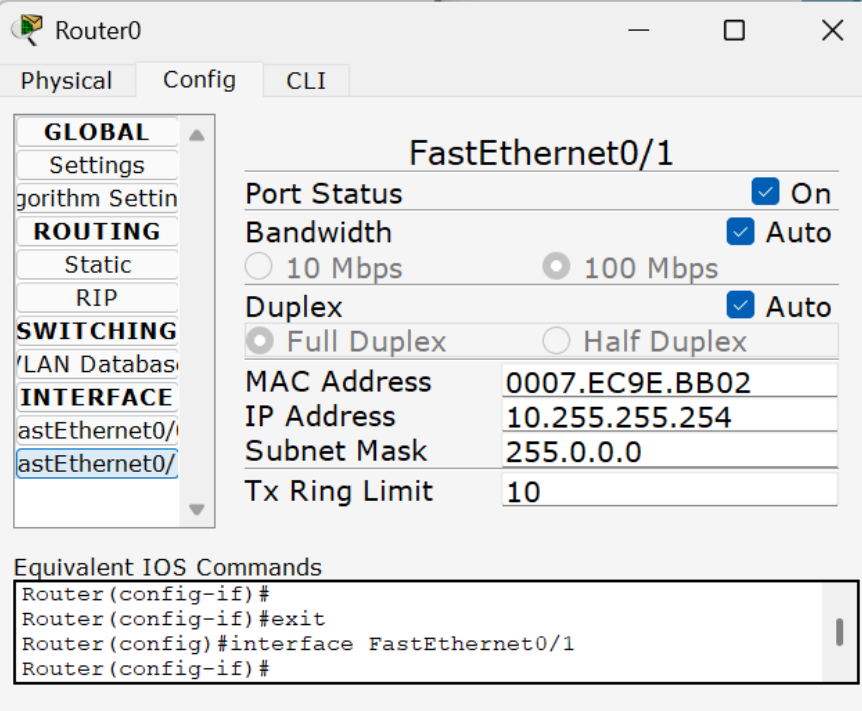


The screenshot shows the configuration window for Router0, specifically for the FastEthernet0/0 interface. The left sidebar contains a tree view with categories: GLOBAL, ROUTING, SWITCHING, and INTERFACE. Under INTERFACE, FastEthernet0/0 is selected. The main area displays the configuration for FastEthernet0/0 with the following settings:

FastEthernet0/0	
Port Status	☒ On
Bandwidth	☒ Auto
	☐ 10 Mbps ☒ 100 Mbps
Duplex	☒ Auto
	☒ Full Duplex ☐ Half Duplex
MAC Address	0007.EC9E.BB01
IP Address	172.16.255.254
Subnet Mask	255.255.0.0
Tx Ring Limit	10

Below the configuration table, there is a section titled "Equivalent IOS Commands" with a text area containing the following commands:

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL
Router(config)#interface FastEthernet0/0
Router(config-if)#
```



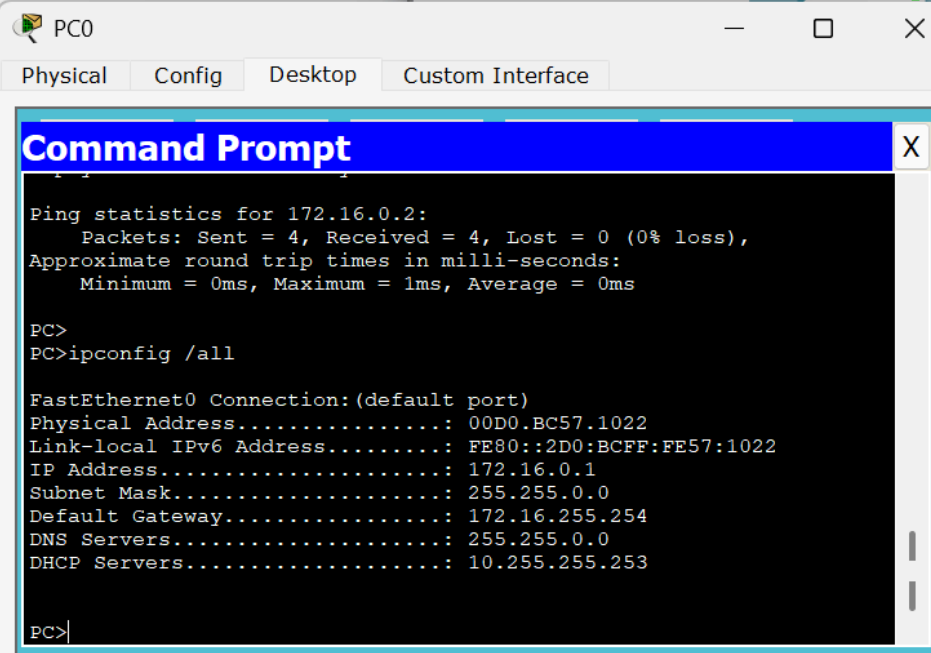
The screenshot shows the configuration window for Router0, specifically for the FastEthernet0/1 interface. The left sidebar is the same as in the first screenshot, with FastEthernet0/1 selected under the INTERFACE category. The main area displays the configuration for FastEthernet0/1 with the following settings:

FastEthernet0/1	
Port Status	☒ On
Bandwidth	☒ Auto
	☐ 10 Mbps ☒ 100 Mbps
Duplex	☒ Auto
	☒ Full Duplex ☐ Half Duplex
MAC Address	0007.EC9E.BB02
IP Address	10.255.255.254
Subnet Mask	255.0.0.0
Tx Ring Limit	10

Below the configuration table, there is a section titled "Equivalent IOS Commands" with a text area containing the following commands:

```
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
```

Test du DHCP sur les clients :



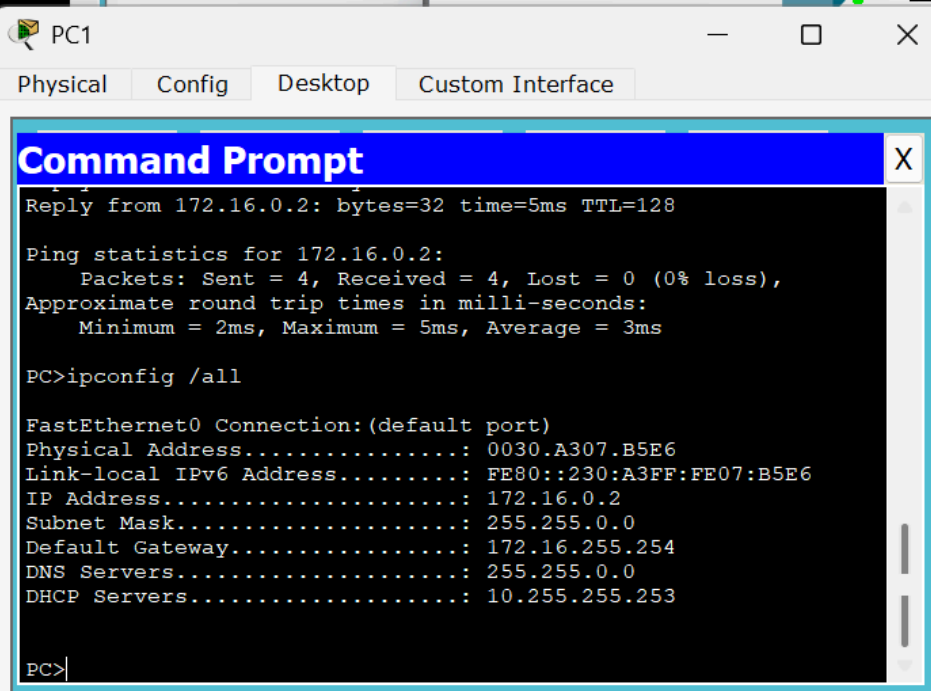
The screenshot shows a window titled 'PC0' with tabs for 'Physical', 'Config', 'Desktop', and 'Custom Interface'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the results of a ping to 172.16.0.2 and the output of the 'ipconfig /all' command.

```
PC0
Physical Config Desktop Custom Interface
Command Prompt
Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
PC>ipconfig /all

FastEthernet0 Connection:(default port)
Physical Address.....: 00D0.BC57.1022
Link-local IPv6 Address.....: FE80::2D0:BCFF:FE57:1022
IP Address.....: 172.16.0.1
Subnet Mask.....: 255.255.0.0
Default Gateway.....: 172.16.255.254
DNS Servers.....: 255.255.0.0
DHCP Servers.....: 10.255.255.253

PC>
```



The screenshot shows a window titled 'PC1' with tabs for 'Physical', 'Config', 'Desktop', and 'Custom Interface'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the results of a ping from 172.16.0.2 and the output of the 'ipconfig /all' command.

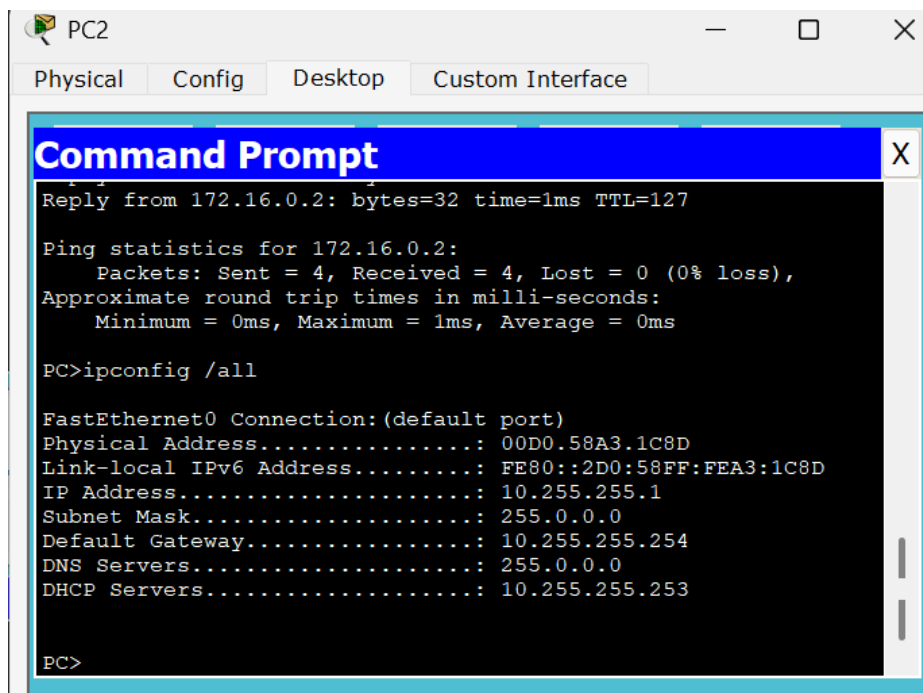
```
PC1
Physical Config Desktop Custom Interface
Command Prompt
Reply from 172.16.0.2: bytes=32 time=5ms TTL=128

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

PC>ipconfig /all

FastEthernet0 Connection:(default port)
Physical Address.....: 0030.A307.B5E6
Link-local IPv6 Address.....: FE80::230:A3FF:FE07:B5E6
IP Address.....: 172.16.0.2
Subnet Mask.....: 255.255.0.0
Default Gateway.....: 172.16.255.254
DNS Servers.....: 255.255.0.0
DHCP Servers.....: 10.255.255.253

PC>
```



PC2

Physical Config Desktop Custom Interface

Command Prompt

```
Reply from 172.16.0.2: bytes=32 time=1ms TTL=127

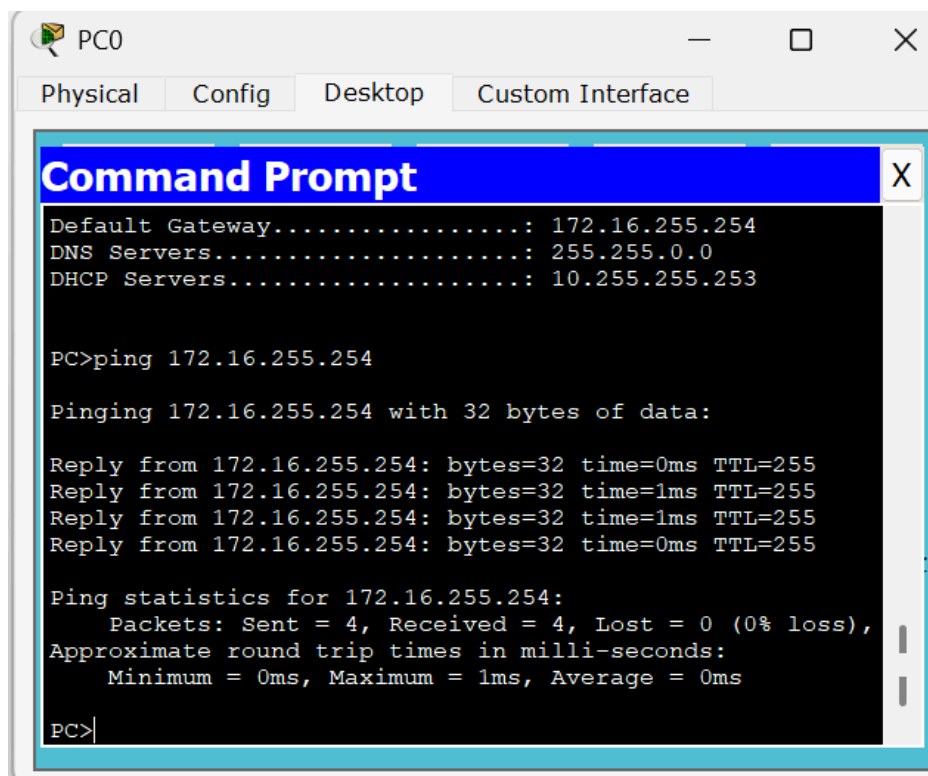
Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ipconfig /all

FastEthernet0 Connection:(default port)
Physical Address.....: 00D0.58A3.1C8D
Link-local IPv6 Address.....: FE80::2D0:58FF:FEA3:1C8D
IP Address.....: 10.255.255.1
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 10.255.255.254
DNS Servers.....: 255.0.0.0
DHCP Servers.....: 10.255.255.253

PC>
```

Test de pings :



PC0

Physical Config Desktop Custom Interface

Command Prompt

```
Default Gateway.....: 172.16.255.254
DNS Servers.....: 255.255.0.0
DHCP Servers.....: 10.255.255.253

PC>ping 172.16.255.254

Pinging 172.16.255.254 with 32 bytes of data:

Reply from 172.16.255.254: bytes=32 time=0ms TTL=255
Reply from 172.16.255.254: bytes=32 time=1ms TTL=255
Reply from 172.16.255.254: bytes=32 time=1ms TTL=255
Reply from 172.16.255.254: bytes=32 time=0ms TTL=255

Ping statistics for 172.16.255.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

PC0

Physical Config Desktop Custom Interface

Command Prompt

```
Ping statistics for 172.16.255.254:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>ping 10.255.255.254

Pinging 10.255.255.254 with 32 bytes of data:

Reply from 10.255.255.254: bytes=32 time=0ms TTL=255
Reply from 10.255.255.254: bytes=32 time=0ms TTL=255
Reply from 10.255.255.254: bytes=32 time=1ms TTL=255
Reply from 10.255.255.254: bytes=32 time=1ms TTL=255

Ping statistics for 10.255.255.254:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

PC0

Physical Config Desktop Custom Interface

Command Prompt

```
Ping statistics for 10.255.255.254:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 1ms, Average = 0ms

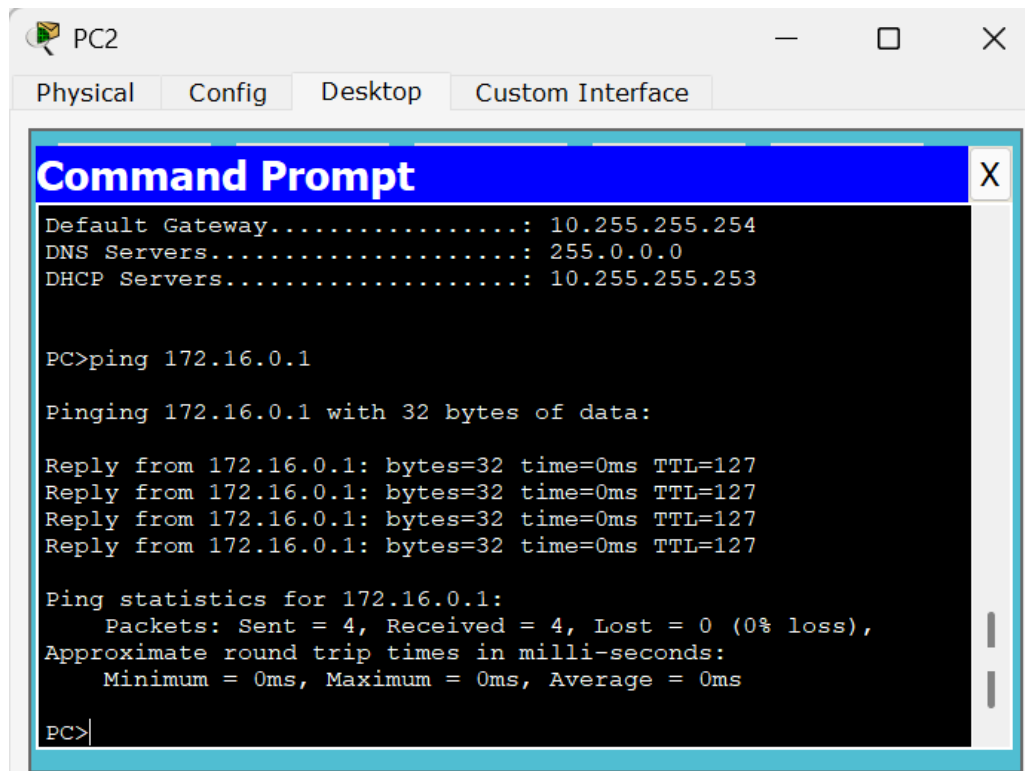
PC>ping 10.255.255.253

Pinging 10.255.255.253 with 32 bytes of data:

Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127
Reply from 10.255.255.253: bytes=32 time=0ms TTL=127

Ping statistics for 10.255.255.253:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>
```



Tous les tests ont permis de vérifier que la configuration marche, le ping vers la passerelle de chaque VLAN répond correctement, ce qui confirme que les interfaces des routeurs sont bien joignables. Les pings entre hôtes de Vlan différents fonctionnent, ce qui prouve que le routage inter vlan marche. Le ping vers le serveur DHCP depuis le vlan20 marche également ce qui montre que l'ip helper transmet bien les requêtes entre les deux réseaux.

Conclusion :

Ce TP visait à séparer le réseau en deux VLAN , VLAN 20 pour les client 0 et 1 et VLAN 10 pour le client 3 et le DHCP, et à assurer l'attribution automatique des adresses IP. Les ports des switches ont été affectés aux VLAN correspondants, puis le routeur a été configuré avec un passerelle pour chaque sous réseau afin de permettre le routage entre eux. Comme le serveur DHCP se trouvait dans un autre réseau l'agent relais DHCP (ip helper-address) a été mis en place pour transmettre les requêtes des clients du VLAN 20 vers le serveur. Les tests ont confirmé que les PC reçoivent une adresse IP correcte et que la communication entre les deux réseaux fonctionne